

LAB SUBMISSION

Build a RAG System from Scratch

Internal policy Q&A: chunking, embeddings, hybrid retrieval, reranking, evaluation, and a planted data-quality diagnosis.

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SECTION 01

Generated source documents

Four generated policies make up the corpus. The planted defect is an archived 2022 copy of the expense policy. It keeps the same section codes as the current policy and disagrees on the figures. The current file is version 3.0 and status current, with a 95 USD international per diem. The archived file is version 1.4 and status archived, with a 75 USD per diem. The same pair also disagrees on the pre-approval threshold: 500 USD now, 1,000 USD in the archive.

<pre>1 ---- 2 title: Expense Reimbursement Policy 3 doc_id: POL-EXP 4 version: "3.0" 5 status: current 6 effective_date: "2026-01-01" 7 owner: Finance Operations 8 supersedes: expense_policy_2022.md 9 ---- 10 11 # Expense Reimbursement Policy 12 13 ## POL-EXP-001 – Purpose and Scope 14 15 This policy governs the reimbursement of business expenses incurred by employees 16 on behalf of the company. It applies to all employees worldwide, including those 17 on hybrid and fully remote arrangements. Expenses must be reasonable, necessary, 18 and directly connected to company business. Where a local statutory requirement 19 conflicts with this policy, the local requirement prevails and Finance 20 Operations will record the exception. 21 22 ## POL-EXP-002 – Submission and Deadlines 23 24 Expense claims must be submitted within 30 calendar days of the date the expense 25 was incurred. Claims submitted after this window require written justification 26 from the employee's manager and are approved at the discretion of Finance 27 Operations. Claims older than 90 calendar days will not be reimbursed under any 28 circumstances. Reimbursements are paid through the standard payroll cycle 29 following approval, and employees should expect payment within two pay periods. 30 31 ## POL-EXP-003 – Receipt Requirements 32 33 An itemised receipt is required for every expense of 25 USD or more. Card 34 statements are not accepted as substitutes for itemised receipts, because they 35 do not evidence what was purchased. For expenses below 25 USD, a written 36 description of the expense and its business purpose is sufficient. Receipts must 37 be legible and must show the vendor, date, amount, and currency. Where a receipt 38 has been lost, employees may submit a missing receipt declaration, limited to 39 three occurrences per calendar year. 40 41 ## POL-EXP-004 – Pre-Approval Thresholds 42 43 Any single expense exceeding 500 USD requires written manager approval before 44 the expense is incurred. Approval must be obtained in advance through the 45 expense system and cannot be granted retrospectively except where the expense 46 arose from a genuine emergency affecting employee safety. Bookings that exceed 47 this threshold and were made without prior approval may be rejected in full, and 48 the cost becomes the personal responsibility of the employee. Recurring 49 subscriptions of any value require pre-approval regardless of the individual 50 charge amount, because the aggregate annual cost is the relevant figure. 51 Departmental budget holders may set lower thresholds for their own teams but may 52 not set higher ones. 53 54 ## POL-EXP-005 – Meals and Entertainment 55 56 Meals taken alone while travelling on company business are reimbursed at actual 57 cost up to the applicable per diem. Meals involving external parties are treated 58 as business entertainment and require the names and employers of all attendees 59 to be recorded on the claim. Alcohol is reimbursable only where it forms part of 60 a client entertainment meal and is limited to two units per attendee. Team 61 social events require prior approval from a departmental budget holder and are 62 capped at 60 USD per attendee per event. 63 64 ## POL-EXP-006 – International Travel and Per Diem 65 66 Employees travelling internationally on company business receive a per diem of 67 95 USD per day to cover meals and incidental expenses. The per diem is paid for 68 each full day of travel and is pro-rated at 50 percent for departure and return</pre>	<pre>1 ---- 2 title: Expense Reimbursement Policy 3 doc_id: POL-EXP 4 version: "1.4" 5 status: archived 6 effective_date: "2022-04-01" 7 superseded_by: expense_policy.md 8 owner: Finance Operations 9 ---- 10 11 # Expense Reimbursement Policy 12 13 ## POL-EXP-001 – Purpose and Scope 14 15 This policy governs the reimbursement of business expenses incurred by employees 16 on behalf of the company. It applies to all permanent employees. Expenses must 17 be reasonable, necessary, and directly connected to company business. Where a 18 local statutory requirement conflicts with this policy, the local requirement 19 prevails and Finance Operations will record the exception. 20 21 ## POL-EXP-002 – Submission and Deadlines 22 23 Expense claims must be submitted within 60 calendar days of the date the expense 24 was incurred. Claims submitted after this window require written justification 25 from the employee's manager and are approved at the discretion of Finance 26 Operations. Claims older than 120 calendar days will not be reimbursed under any 27 circumstances. Reimbursements are paid by separate bank transfer twice monthly 28 following approval, and employees should allow up to three weeks from approval 29 for funds to clear. Claims spanning more than one calendar month should be 30 submitted separately so that each month can be allocated to the correct 31 accounting period. 32 33 ## POL-EXP-003 – Receipt Requirements 34 35 An itemised receipt is required for every expense of 75 USD or more. Card 36 statements may be accepted as supporting evidence where an itemised receipt is 37 not available. For expenses below 75 USD, a written description of the expense 38 and its business purpose is sufficient. Receipts must be legible and must show 39 the vendor, date, amount, and currency. Handwritten receipts are accepted where 40 a vendor does not issue printed documentation, provided the vendor name and 41 address are shown. Foreign language receipts should be accompanied by a short 42 English summary of the items purchased. 43 44 ## POL-EXP-004 – Pre-Approval Thresholds 45 46 Any single expense exceeding 1,000 USD requires written manager approval before 47 the expense is incurred. Approval should normally be obtained in advance, but 48 may be granted retrospectively where the employee provides a business 49 justification within five working days. Recurring subscriptions above 1,000 USD 50 annually require pre-approval. Departmental budget holders may set their own 51 thresholds for their teams. 52 53 ## POL-EXP-005 – Meals and Entertainment 54 55 Meals taken alone while travelling on company business are reimbursed at actual 56 cost up to the applicable per diem. Meals involving external parties are treated 57 as business entertainment and require the names and employers of all attendees 58 to be recorded on the claim. Alcohol is reimbursable where it forms part of a 59 client entertainment meal. Team social events require prior approval from a 60 departmental budget holder and are capped at 40 USD per attendee per event. 61 62 ## POL-EXP-006 – International Travel and Per Diem 63 64 Employees travelling internationally on company business receive a per diem of 65 75 USD per day to cover meals and incidental expenses. The per diem is paid for 66 each full day of travel and is pro-rated at 50 percent for departure and return 67 days. Receipts are not required against the per diem, as it replaces itemised 68 meal claims for the duration of the trip. Lodging is reimbursed separately at</pre>
---	---

Both expense policies, including version, status, and the conflicting per-diem lines.

1	social events require prior approval from a	60	require prior approval from a
2	departmental budget holder and are	61	departmental budget holder and are capped at 40
3	capped at 60 USD per attendee per event.	62	USD per attendee per event.
4	## POL-EXP-006 – International Travel and Per	63	## POL-EXP-006 – International Travel and Per
5	Diem	64	Diem
6	Employees travelling internationally on company	65	Employees travelling internationally on company
7	business receive a per diem of	66	business receive a per diem of
8	95 USD per day to cover meals and incidental	67	75 USD per day to cover meals and incidental
9	expenses. The per diem is paid for	68	expenses. The per diem is paid for
10	each full day of travel and is pro-rated at 50		each full day of travel and is pro-rated at 50
11	percent for departure and return		percent for departure and return
12	days. Receipts are not required against the per		days. Receipts are not required against the per
13	diem, as it replaces itemised		diem, as it replaces itemised
14			meal claims for the duration of the trip.

Close view of POL-EXP-006: 95 USD in the current policy and 75 USD in the 2022 archive.

```
● ((.venv) ) root@f830357d28a3:/workspace/rag_policy_assistant# ls data/raw
expense_policy.md expense_policy_2022.md pto_policy.md remote_work_policy.md
○ ((.venv) ) root@f830357d28a3:/workspace/rag_policy_assistant# █
```

All four source filenames in data/raw.

SECTION 02

Minimal embed, store, and retrieve loop

Before the full pipeline, two known snippets were embedded with sentence-transformers and stored in a throwaway Chroma collection. The query is a paraphrase, not a copy of either snippet. The per-diem snippet ranks first, which confirms that embedding, storage, and vector search work.

```
● ((.venv) ) root@f830357d28a3:/workspace/rag_policy_assistant# python scripts/minimal_loop.py
minimal embed-store-retrieve loop
query: What is the meal allowance for overseas travel?
```

rank	chunk_id	cosine distance	closer?
1	snippet::per_diem::0	0.5964	YES <-- expected
2	snippet::pto::0	0.8046	no

PASS: paraphrased query retrieved the per-diem snippet first.
cosine distance: lower is closer (hnspace=cosine).

```
○ ((.venv) ) root@f830357d28a3:/workspace/rag_policy_assistant# █
```

Paraphrased query ranks snippet::per_diem::0 first. Lower cosine distance is closer.

SECTION 03

Chunking, embeddings, and the vector store

Documents are split on policy-section headings, then windowed at 160 words with a 32-word overlap only when a section overflows. `store.py` persists chunks in Chroma with cosine distance. `chunking.py` rejects a chunk that would exceed the embedding model's token ceiling or whose character span does not match the raw file. `embeddings.py` loads a `SentenceTransformer`, counts tokens with that model's tokenizer, and encodes the chunk text. The ingest run produced 4 documents, 26 sections, and 27 chunks. The longest chunk is 203 tokens, under the 256-token ceiling.

```

rag_policy_assistant > src > rag > store.py > VectorStore > __init__
8
9
10 > PersistentClient Aa Ab, a* 1 of 1 ↑ ↓ ≡ ×
11 rag.types import Chunk
12
13
14 aclass(frozen=True)
15 s DenseHit:
16 chunk_id: str
17 text: str
18 distance: float
19 metadata: dict[str, Any]
20
21
22 s VectorStore:
23 def __init__(self, persist_dir: Path, collection: str) -> None:
24     self.persist_dir.mkdir(parents=True, exist_ok=True)
25     self._client = chromadb.PersistentClient(path=str(persist_dir))
26     self._collection_name = collection
27     self._collection = self._client.get_or_create_collection(
28         name=collection,
29         metadata={"hnsw:space": "cosine"},
30     )
31
32 def reset(self) -> None:
33     """Drop and recreate so every ingest starts from an ident
34     try:
35         self._client.delete_collection(self._collection_name)
36     except Exception:
37         pass
38     self._collection = self._client.get_or_create_collection(
39         name=self._collection_name,
40         metadata={"hnsw:space": "cosine"},
41     )
42
43 def upsert(self, chunks: list[Chunk], embeddings: list[list[float]]) -> None:
44     if len(chunks) != len(embeddings):
45         raise ValueError("chunks and embeddings must be the same length")
46     self._collection.upsert(
47         ids=[chunk.chunk_id for chunk in chunks],
48         embeddings=cast("list[Sequence[float]]", embeddings),
49         documents=[chunk.embed_text() for chunk in chunks],
50         metadatas=[chunk.metadata() for chunk in chunks],
51     )
52     return len(chunks)

```

```

rag_policy_assistant > src > rag > chunking.py > word_histogram
100
101
102 > 160 Aa Ab, a* 2 of 2 ↑ ↓ ≡ ×
103 _title",
104
105 chunks:
106 d in required:
107     ot str(getattr(chunk, field)).strip():
108         errors.append(f"{chunk.chunk_id}: missing {field}")
109     hunk.text.strip():
110         rs.append(f"{chunk.chunk_id}: empty text")
111     .token_count > max_tokens:
112         rs.append(
113             f"{chunk.chunk_id}: {chunk.token_count} tokens exceeds "
114             f"model ceiling of {max_tokens} (silent truncation)"
115         )
116
117 by_path.get(chunk.doc)
118 e is None:
119     rs.append(f"{chunk.chunk_id}: unknown source document {chunk.source}")
120
121 source.raw[chunk.char_start:chunk.char_end]
122 d = chunk.text:
123     rs.append(
124         f"{chunk.chunk_id}: span {chunk.char_start}:{chunk.char_end} "
125         f"does not round-trip against the raw file"
126     )
127
128 "\n ".join(errors)
129 unKValidationError(f"{len(errors)} chunk validation error(s): {errors}")
130
131
132 am(chunks: list[Chunk]) -> list[tuple[str, int]]:
133     rd counts so the size choice is visible, not implicit."
134     80, 120, 160, 200, 10_000
135     40, "41-80", "81-120", "121-160", "161-200", "201+"
136     * len(edges)
137     chunks:
138     x, edge in enumerate(Literal[40, 80, 120, 160, 200, 10000]):
139         hunk.word_count <= edge:
140             counts[index] += 1
141         break
142     tuple[str, int](zip(tuple[str, int](labels, counts))

```

```

rag_policy_assistant > src > rag > embeddings.py > ...
1
2 """Cached SentenceTransformer loader, token counting, and embedding"""
3 from __future__ import annotations
4
5 from functools import lru_cache
6
7 from sentence_transformers import SentenceTransformer
8
9 from config import EmbeddingModel, active_embedding_model
10
11 @lru_cache(maxsize=4)
12 def load_model(model_id: str) -> SentenceTransformer:
13     return SentenceTransformer(model_id)
14
15
16 def current_model() -> tuple[EmbeddingModel, SentenceTransformer]:
17     spec = active_embedding_model()
18     return spec, load_model(spec.model_id)
19
20
21 def count_tokens(text: str, model: SentenceTransformer | None) -> int:
22     """Count wordpieces the same way the embedding model will"""
23     if model is None:
24         model = current_model()
25     encoded = model.tokenizer(text, add_special_tokens=True,
26                               return_ints=True)
27     return len(encoded["input_ids"])
28
29
30 def encode_texts(texts: list[str], model: SentenceTransformer | None) -> list[list[float]]:
31     if model is None:
32         model = current_model()
33     vectors = model.encode(
34         texts,
35         normalize_embeddings=True,
36         show_progress_bar=False,
37     )
38     return [vector.tolist() for vector in vectors]
39
40
41 def encode_query(query: str, model: SentenceTransformer | None) -> list[float]:
42     spec = active_embedding_model()
43     prefixed = f"{spec.query_prefix}{query}" if spec.query_prefix else query
44     return encode_texts([prefixed], model=model)[0]

```

`store.py`, `chunking.py`, and `embeddings.py`: Chroma cosine storage, chunk validation, and `SentenceTransformer` encoding.

```

● ((.venv) ) root@f830357d28a3:/workspace/rag_policy_assistant# python scripts/ingest.py
embedding model : sentence-transformers/all-MiniLM-L6-v2
token ceiling   : 256
window / overlap: 160 / 32 words
documents       : 4
sections        : 26
chunks          : 27

```

chunk word-count histogram

```

-----
1-40      0
41-80     10 #####
81-120    14 #####
121-160    3 ###
161-200    0
201+      0

```

tokens min=78 mean=121.8 max=203 ceiling=256

windowed sections: 1

stored in chroma : 27 vectors at /workspace/rag_policy_assistant/data/chroma

Ingest output: 160/32 window, 27 chunks, maximum 203 tokens, stored in Chroma.

SECTION 04

Basic RAG pipeline, end to end

A natural-language question is embedded, the top chunks are retrieved, and Ollama mistral:7b writes the answer at temperature 0. The answer states the current international per diem, 95 USD, and cites expense_policy.md, section POL-EXP-006, version 3.0, status current.

```

• ((.venv) ) root@f830357d28a3:/workspace/rag_policy_assistant# python scripts/ask.py --mode hybrid \
  "What is the international per diem for meals and incidentals?"
Q: What is the international per diem for meals and incidentals?

The international per diem for meals and incidentals is 95 USD per day for employees traveling internationally on company business.
The per diem is paid for each full day of travel and is pro-rated at 50 percent for departure and return days. Receipts are not re
quired against the per diem. [S1]

model: mistral:7b  retrieval: hybrid
sources:
  [S1] expense_policy.md § POL-EXP-006 "International Travel and Per Diem" (v3.0, current, chars 3106-3810) dist=0.967
  [S2] expense_policy.md § POL-EXP-005 "Meals and Entertainment" (v3.0, current, chars 2537-3052) dist=0.969
  [S3] expense_policy.md § POL-EXP-001 "Purpose and Scope" (v3.0, current, chars 250-681) dist=0.970
○ ((.venv) ) root@f830357d28a3:/workspace/rag_policy_assistant#

```

Hybrid retrieval followed by a grounded answer and a source list.

SECTION 05

Hybrid retrieval beats vector-only search

keyword.py matches exact POL- codes, with a large score boost, and also counts ordinary term overlap. hybrid.py combines that ranked list with dense search using reciprocal rank fusion, $1/(k + \text{rank})$. The code-only query POL-EXP-004 is the demonstration: dense search ranks the section 11th, outside the top five. Hybrid search ranks it 4th.

```
rag_policy_assistant > src > rag > keyword.py > ...
26 def normalize(text: str) -> str:
27     return NON_TOKEN.sub(" ", text.casefold()).strip()
28
29
30 def terms(text: str) -> list[str]:
31     return [token for token in normalize(text).split() if token and token not in STOPWORDS]
32
33
34 def extract_codes(text: str) -> list[str]:
35     return [match.group(0).upper() for match in CODE.finditer(text)]
36
37
38 def _score(query: str, hit: DenseHit) -> float:
39     codes = extract_codes(query)
40     haystack = f"{hit.text} {hit.metadata.get('section_code', '')}".upper()
41     score = 0.0
42     for code in codes:
43         if code in haystack:
44             score += 100.0
45     query_terms = terms(query)
46     if not query_terms:
47         return score
48     blob = normalize(hit.text)
49     section = normalize(str(hit.metadata.get("section_title", "")))
50     blob = f"{blob} {section}"
51     overlap = sum(1 for term in query_terms if term in blob)
52     score += float(overlap)
53     return score
54
55
56 def keyword_search(query: str, corpus: list[DenseHit], k: int) -> list[KeywordHit]:
57     scored: list[KeywordHit] = []
58     for hit in corpus:
59         value = _score(query, hit)
60         if value <= 0:
61             continue
62         scored.append(
63             KeywordHit(
64                 chunk_id=hit.chunk_id,
65                 text=hit.text,
66                 score=value,
67                 metadata=dict(str, object)(hit.metadata),
68             )
69         )
70     scored.sort(key=lambda item: (-item.score, item.chunk_id))
71     return scored[:k]

rag_policy_assistant > src > rag > hybrid.py > FusedHit
...
12
13 @dataclass(frozen=True)
14 class FusedHit:
15     chunk_id: str
16     text: str
17     metadata: dict[str, object]
18     rrf: float
19     dense_rank: int | None
20     keyword_rank: int | None
21
22
23 def reciprocal_rank_fusion(
24     rankings: list[list[str]],
25     k: int = RRF_K,
26 ) -> dict[str, float]:
27     """score(id) = sum 1/(k + rank) over lists that contain the id. Ranks are 1-based."""
28     scores: dict[str, float] = {}
29     for ranking in rankings:
30         for rank, chunk_id in enumerate(ranking, start=1):
31             scores[chunk_id] = scores.get(chunk_id, 0.0) + 1.0 / (k + rank)
32     return scores
33
34
35 def fuse(
36     dense: list[DenseHit],
37     keyword: list[KeywordHit],
38     *,
39     rrf_k: int = RRF_K,
40 ) -> list[FusedHit]:
41     dense_ids = {hit.chunk_id for hit in dense}
42     keyword_ids = {hit.chunk_id for hit in keyword}
43     scores = reciprocal_rank_fusion([dense_ids, keyword_ids], k=rrf_k)
44     dense_rank = {hit.chunk_id: index for index, hit in enumerate(dense, start=1)}
45     keyword_rank = {hit.chunk_id: index for index, hit in enumerate(keyword, start=1)}
46     by_id: dict[str, tuple[str, Mapping[str, object]]] = {}
47     for dense_hit in dense:
48         by_id[dense_hit.chunk_id] = (dense_hit.text, dense_hit.metadata)
49     for keyword_hit in keyword:
50         by_id.setdefault(keyword_hit.chunk_id, (keyword_hit.text, keyword_hit.metadata))
51     fused = []
52     for chunk_id in dense_ids | keyword_ids:
53         fused_hit = FusedHit(
54             chunk_id=chunk_id,
55             text=by_id[chunk_id][0],
56             metadata=dict(str, object)(by_id[chunk_id][1]),
57             rrf=scores[chunk_id],
58         )
59         fused.append(fused_hit)
60     return fused
```

Keyword code matching and reciprocal rank fusion.

```
• ((.venv) ) root@f830357d28a3:/workspace/rag_policy_assistant# python scripts/compare_retrieval.py --query POL-EXP-004
corpus size: 27 candidate_k: 20
```

```
Q: POL-EXP-004
target POL-EXP-004
dense=11 keyword=1 hybrid=4 rerank=1
dense top5 : POL-EXP-001, POL-EXP-001, POL-EXP-003, POL-EXP-007, POL-EXP-002
hybrid top5 : POL-EXP-001, POL-EXP-002, POL-EXP-001, POL-EXP-004, POL-EXP-007
rerank top5 : POL-EXP-004, POL-EXP-004, POL-EXP-007, POL-EXP-001, POL-EXP-007
rerank #1 p=1.000 logit=+8.08 expense_policy_2022.md:POL-EXP-004:0
DEMO: vector-only misses the LLM top-5; hybrid recovers it. This is the locked hybrid-wins query.
```

```
o ((.venv) ) root@f830357d28a3:/workspace/rag_policy_assistant#
```

dense=11, keyword=1, hybrid=4, with the DEMO line for the locked hybrid-wins query.

SECTION 06

Cross-encoder reranking

After the first retrieval pass, a cross-encoder scores each query and chunk together. The model is cross-encoder/ms-marco-MiniLM-L-6-v2. Logits are converted with a sigmoid, then the top candidates are kept. On POL-EXP-004, reranking moves the target section to rank 1.

```
rag_policy_assistant > src > rag > rerank.py > _default_scorer

31 def sigmoid(logit: float) -> float:
32     if logit >= 0:
33         return 1.0 / (1.0 + math.exp(-logit))
34     exp = math.exp(logit)
35     return exp / (1.0 + exp)
36
37
38 @lru_cache(maxsize=2)
39 def load_reranker(model_id: str = RERANK_MODEL) -> CrossEncoder:
40     return cast(CrossEncoder, CrossEncoder(model_id))
41
42
43 def _default_scorer(pairs: Sequence[Pair]) -> Sequence[float]:
44     model = load_reranker()
45     scores = model.predict(list[Pair](pairs), show_progress_bar=False)
46     return [float(score) for score in scores]
47
48
49 def rerank(
50     query: str,
51     hits: Sequence[FusedHit | DenseHit | RerankedHit],
52     *,
53     top_k: int,
54     min_prob: float = MIN_RERANK_PROB,
55     scorer: Scorer | None = None,
56 ) -> list[RerankedHit]:
57     """Re-score candidates. Query is always the first member of each pair."""
58     if not hits:
59         return []
60     score_fn = scorer or _default_scorer
61     pairs: list[Pair] = [(query, hit.text) for hit in hits]
62     logits = list[float](score_fn(pairs))
```

```
Problems 1 Output Debug Console Terminal Ports
bash - rag_policy_assistant + v [ ] [ ] ...

[S1] expense_policy.md $ POL-EXP-006 "International Travel and Per Diem" (v3.0, current, chars 3106-3810) dist=0.967
[S2] expense_policy.md $ POL-EXP-005 "Meals and Entertainment" (v3.0, current, chars 2537-3052) dist=0.969
[S3] expense_policy.md $ POL-EXP-001 "Purpose and Scope" (v3.0, current, chars 250-681) dist=0.970
• ((.venv) ) root@f830357d28a3:/workspace/rag_policy_assistant# python scripts/compare_retrieval.py --query POL-EXP-004
corpus size: 27 candidate_k: 20

Q: POL-EXP-004
target POL-EXP-004
dense=11 keyword=1 hybrid=4 rerank=1
dense top5 : POL-EXP-001, POL-EXP-001, POL-EXP-003, POL-EXP-007, POL-EXP-002
hybrid top5 : POL-EXP-001, POL-EXP-002, POL-EXP-001, POL-EXP-004, POL-EXP-007
rerank top5 : POL-EXP-004, POL-EXP-004, POL-EXP-007, POL-EXP-001, POL-EXP-007
rerank #1 p=1.000 logit=+8.08 expense_policy_2022.md::POL-EXP-004::0
DEMO: vector-only misses the LLM top-5; hybrid recovers it. This is the locked hybrid-wins query.
○ ((.venv) ) root@f830357d28a3:/workspace/rag_policy_assistant#
```

rerank.py pair scoring, and the same query with rerank=1.

SECTION 07

Evaluation set and harness

The fixed set has 11 cases: 10 answerable questions and one refusal. Each answerable case names the expected section, document, and keywords. Retrieval tests measure recall at 5 and mean reciprocal rank. Generation tests check that the answer contains the expected facts and that every [S#] citation points at a retrieved source.

```
rag_policy_assistant > tests > ! eval_set.yaml
```

```

1  cases:
2    - id: per_diem
3      question: What is the international per diem for meals and incidentals?
4      expected_section: POL-EXP-006
5      expected_doc: expense_policy.md
6      expected_keywords:
7        - "95"
8      retrieval: hybrid
9
10   - id: pre_approval
11     question: What is the pre-approval threshold for a single expense?
12     expected_section: POL-EXP-004
13     expected_doc: expense_policy.md
14     expected_keywords:
15       - "500"
16     retrieval: hybrid
17
18   - id: expense_deadline
19     question: How many days do I have to submit an expense claim?
20     expected_section: POL-EXP-002
21     expected_doc: expense_policy.md
22     expected_keywords:
23       - "30"
24     retrieval: hybrid
25
26   - id: receipt_threshold
27     question: From what amount do I need an itemised receipt?
28     expected_section: POL-EXP-003
29     expected_doc: expense_policy.md
30     expected_keywords:
31       - "25"
32     retrieval: hybrid
33
34   - id: pto_carryover
35     question: How many unused PTO days can I carry into the next year?
36     expected_section: POL-PTO-003
37     expected_doc: pto_policy.md
38     expected_keywords:
39       - "5"
40     retrieval: hybrid
41
```

First half of tests/eval_set.yaml, with expected sections and keywords.

```
rag_policy_assistant > tests > ! eval_set.yaml
```

```

1  cases:
50 - id: parental_leave
51   question: How many weeks of paid parental leave are employees entitled to?
52   expected_section: POL-PTO-005
53   expected_doc: pto_policy.md
54   expected_keywords:
55     - "16"
56   retrieval: hybrid
57
58 - id: core_hours
59   question: What are the core collaboration hours for remote employees?
60   expected_section: POL-RMT-002
61   expected_doc: remote_work_policy.md
62   expected_keywords:
63     - "10:00"
64     - "16:00"
65   retrieval: hybrid
66
67 - id: stipend
68   question: What is the one-time home office stipend for remote employees?
69   expected_section: POL-RMT-003
70   expected_doc: remote_work_policy.md
71   expected_keywords:
72     - "800"
73   retrieval: hybrid
74
75 - id: policy_code
76   question: POL-EXP-004
77   expected_section: POL-EXP-004
78   expected_doc: null
79   expected_keywords:
80     - approval
81   retrieval: hybrid
82   notes: Code-only query. Dense misses top-5; hybrid recovers a POL-EXP-004 chunk.
83
84 - id: tuition_refusal
85   question: What is the annual tuition reimbursement cap for employees?
86   expected_section: null
87   expected_doc: null
88   expected_keywords: []
89   retrieval: hybrid
90   expect_refusal: true

```

Remaining cases, including the code-only query and the tuition refusal.

```

rag_policy_assistant > tests > test_retrieval_eval.py > ...
33
34 def test_hybrid_recall_and_mrr(
35     vector_store: VectorStore,
36     answerable_cases: list[dict[str, Any]],
37 ) -> None:
38     pipeline = RAGPipeline(store=vector_store)
39     ranks: list[int | None] = []
40     print("\nhybrid retrieval")
41     for case in answerable_cases:
42         sources = pipeline.retrieve(case["question"], k=TOP_K, mode="hybrid")
43         rank = _rank(sources, case)
44         ranks.append(rank)
45         print(f" {case['id']:<18} rank={rank or 'miss'}")
46     recall, mrr = _metrics(ranks)
47     print(f" recall@{TOP_K}={recall:.2f} MRR={mrr:.2f}")
48     assert recall >= 0.80
49     assert mrr >= 0.70
50
51
52 def test_hybrid_beats_dense_on_policy_code_query(vector_store: VectorStore) -> None:
53     """Code-only query: dense misses POL-EXP-004; keyword recovers a copy of it.
54
55     Two versions of the section exist (current v3.0 and planted v1.4). The
56     hybrid win is recovering the section code into top-5, not picking the
57     current file – the stale copy is a Phase 7 diagnosis issue.
58     """
59     pipeline = RAGPipeline(store=vector_store)
60     question = "POL-EXP-004"
61     case = {"expected_section": "POL-EXP-004"}
62     dense_rank = _rank(pipeline.retrieve(question, k=20, mode="dense"), case)
63     hybrid_rank = _rank(pipeline.retrieve(question, k=5, mode="hybrid"), case)
64     assert dense_rank is None or dense_rank > 5, dense_rank
65     assert hybrid_rank is not None and hybrid_rank <= 5, hybrid_rank
66
67
68 def test_gold_keywords_fit_in_one_expected_chunk(
69     vector_store: VectorStore,
70     answerable_cases: list[dict[str, Any]],
71 ) -> None:
72     corpus = vector_store.get_all()
73     for case in answerable_cases:
74         keywords = [str(item).lower() for item in case.get("expected_keywords") or []]

```

Retrieval harness: recall@5 and MRR gates, plus the hybrid-versus-dense assertion.

```

rag_policy_assistant > tests > test_generation_eval.py > ...
31 def _load_fixtures() -> dict[str, Any]:
32     """
33     pytest.skip("no generated_answers.json; run scripts/record_answers.py")
34     payload: dict[str, Any] = json.loads(FIXTURE.read_text(encoding="utf-8"))
35     return payload
36
37
38 def test_fixture_answers_cover_key_facts_and_valid_citations(
39     eval_cases: list[dict[str, Any]],
40 ) -> None:
41     payload = _load_fixtures()
42     answers = payload["answers"]
43     missing = [case["id"] for case in eval_cases if case["id"] not in answers]
44     assert not missing, f"fixtures missing cases: {missing}"
45
46     for case in eval_cases:
47         record = answers[case["id"]]
48         text = str(record["text"]).lower()
49         markers = {int(item) for item in MARKER.findall(record["text"])}
50         n_sources = int(record.get("n_sources") or 0)
51         if case.get("expect_refusal"):
52             assert any(hint in text for hint in REFUSAL_HINTS), case["id"]
53             continue
54         for keyword in case.get("expected_keywords") or []:
55             assert str(keyword).lower() in text, f"{case['id']} missing {keyword!r}"
56         assert markers, f"{case['id']} has no [S#] citations"
57         assert all(1 <= marker <= n_sources for marker in markers), case["id"]
58
59
60 @pytest.mark.live_llm
61 def test_live_generation_matches_key_facts(
62     vector_store: VectorStore,
63     eval_cases: list[dict[str, Any]],
64 ) -> None:
65     if not ollama_available():
66         pytest.skip("Ollama is not reachable")
67     pipeline = RAGPipeline(store=vector_store)
68     for case in eval_cases:
69         answer = pipeline.answer(
70             case["question"],
71             k=5,
72             mode=str(case.get("retrieval") or "hybrid"),
73         )
74         text = answer.text.lower()
75         if case.get("expect_refusal"):
76             assert any(hint in text for hint in REFUSAL_HINTS)
77             continue
78         for keyword in case.get("expected_keywords") or []:
79             assert str(keyword).lower() in text
80         cited = {int(item) for item in MARKER.findall(answer.text)}
81         assert cited
82         assert all(1 <= marker <= len(answer.sources) for marker in cited)
83
84

```

Answer-accuracy harness: expected keywords and valid citation markers.

SECTION 08

Evaluation results

The harness was run with pytest. Hybrid retrieval reached the expected chunk for every answerable case. Recall at 5 is 1.00 and mean reciprocal rank is 0.93. The generation check against the recorded answers also passed.

```

DEMO: vector-only misses the LLM top-5; hybrid recovers it. This is the locked hybrid-wins query.
• ((.venv) ) root@f830357d28a3:/workspace/rag_policy_assistant# pytest -s tests/test_retrieval_eval.py tests/test_
  _generation_eval.py
===== test session starts =====
platform linux -- Python 3.12.14, pytest-8.4.2, pluggy-1.6.0 -- /workspace/rag_policy_assistant/.venv/bin/pytho
n3
cachedir: .pytest_cache
rootdir: /workspace/rag_policy_assistant
configfile: pytest.ini
plugins: anyio-4.15.1
collected 5 items / 1 deselected / 4 selected

tests/test_retrieval_eval.py::test_hybrid_recall_and_mrr
hybrid retrieval
  per_diem          rank=1
  pre_approval      rank=1
  expense_deadline  rank=1
  receipt_threshold rank=1
  pto_carryover     rank=1
  sick_leave        rank=1
  parental_leave    rank=1
  core_hours        rank=1
  stipend           rank=1
  policy_code       rank=4
  recall@5=1.00  MRR=0.93
PASSED
tests/test_retrieval_eval.py::test_hybrid_beats_dense_on_policy_code_query PASSED
tests/test_retrieval_eval.py::test_gold_keywords_fit_in_one_expected_chunk PASSED
tests/test_generation_eval.py::test_fixture_answers_cover_key_facts_and_valid_citations PASSED

```

recall@5=1.00, MRR=0.93, and all selected evaluation tests passed.

SECTION 09

Planted data-quality issue and diagnosis

The question is the international per diem. Retrieval is healthy: hybrid search returns the current section first and the archived copy second, and reranking still keeps both in the top results. The raw files, before any embedding, say 95 USD in the current policy and 75 USD in the 2022 archive. The model is therefore handed two contradictory figures. A second planted conflict is the pre-approval threshold, 500 USD now and 1,000 USD in the archive. One unfiltered answer states either 1,000 USD or 500 USD. The failure traces to the source data, not to retrieval or generation.

```

• ((.venv) ) root@f830357d28a3:/workspace/rag_policy_assistant# python scripts/diagnose.py
=====
corpus-wide version collisions (every stored chunk, no query)
=====
LINEAGE WARNING: 7 section(s) retrieved in more than one document version. Archived copies were not dropped.
POL-EXP-001 versions 1.4, 3.0
  [S1] expense_policy.md v3.0 current
  [S8] expense_policy_2022.md v1.4 archived
POL-EXP-002 versions 1.4, 3.0
  [S2] expense_policy.md v3.0 current
  [S9] expense_policy_2022.md v1.4 archived
POL-EXP-003 versions 1.4, 3.0
  [S3] expense_policy.md v3.0 current
  [S10] expense_policy_2022.md v1.4 archived
POL-EXP-004 versions 1.4, 3.0
  [S4] expense_policy.md v3.0 current
  [S11] expense_policy_2022.md v1.4 archived
POL-EXP-005 versions 1.4, 3.0
  [S5] expense_policy.md v3.0 current
  [S12] expense_policy_2022.md v1.4 archived
POL-EXP-006 versions 1.4, 3.0
  [S6] expense_policy.md v3.0 current
  [S13] expense_policy_2022.md v1.4 archived
POL-EXP-007 versions 1.4, 3.0
  [S7] expense_policy.md v3.0 current
  [S14] expense_policy_2022.md v1.4 archived
This is a corpus defect: the retriever returned contradictory policy. Prefer status=current, or remove the stale document from the source system.

=====
case: per_diem
Q: What is the international per diem for meals and incidentals?
=====

1. retrieval – was the gold chunk found?
  hybrid
  [S1] POL-EXP-006 v3.0 current expense_policy.md dist=0.967 <-- gold section
  [S2] POL-EXP-006 v1.4 archived expense_policy_2022.md dist=0.968 <-- gold section
  [S3] POL-EXP-005 v3.0 current expense_policy.md dist=0.969
  [S4] POL-EXP-005 v1.4 archived expense_policy_2022.md dist=0.969
  [S5] POL-EXP-001 v3.0 current expense_policy.md dist=0.970

  LINEAGE WARNING: 2 section(s) retrieved in more than one document version. Archived copies were not dropped
  .
  POL-EXP-005 versions 1.4, 3.0
    [S3] expense_policy.md v3.0 current
    [S4] expense_policy_2022.md v1.4 archived
  POL-EXP-006 versions 1.4, 3.0
    [S1] expense_policy.md v3.0 current
    [S2] expense_policy_2022.md v1.4 archived
  This is a corpus defect: the retriever returned contradictory policy. Prefer status=current, or remove the stale document from the source system.
  rerank
  [S1] POL-EXP-006 v1.4 archived expense_policy_2022.md dist=0.002 <-- gold section
  [S2] POL-EXP-006 v3.0 current expense_policy.md dist=0.003 <-- gold section
  [S3] POL-EXP-005 v1.4 archived expense_policy_2022.md dist=0.491

```

Both versions of POL-EXP-006 are retrieved. The current chunk is rank 1 on the hybrid path.

2. source data – grepped on disk, before any embedding
 expense_policy.md:66: per diem of 95 USD per day
 expense_policy_2022.md:64: per diem of 75 USD per day

3. generation with DETECT_CONFLICTS off (hybrid, eval path)
 Q: What is the international per diem for meals and incidentals?

The international per diem for meals and incidental expenses is 95 USD per day, as stated in the current version of POL-EXP-006 [S1]. The per diem is paid for each full day of travel and is pro-rated at 50 percent for departure and return days. Receipts are not required against the per diem. Lodging is reimbursed separately at actual cost, capped at 280 USD per night in tier-one cities and 190 USD per night elsewhere [S1].

model: mistral:7b retrieval: hybrid
 sources:

[S1] expense_policy.md § POL-EXP-006 "International Travel and Per Diem" (v3.0, current, chars 3106–3810) dist=0.967
 [S2] expense_policy_2022.md § POL-EXP-006 "International Travel and Per Diem" (v1.4, archived, chars 2954–3513) dist=0.968
 [S3] expense_policy.md § POL-EXP-005 "Meals and Entertainment" (v3.0, current, chars 2537–3052) dist=0.969
 [S4] expense_policy_2022.md § POL-EXP-005 "Meals and Entertainment" (v1.4, archived, chars 2431–2900) dist=0.969
 [S5] expense_policy.md § POL-EXP-001 "Purpose and Scope" (v3.0, current, chars 250–681) dist=0.970

3b. generation with DETECT_CONFLICTS off (rerank, unfiltered)
 Q: What is the international per diem for meals and incidentals?

The international per diem for meals and incidental expenses is \$95 per day for employees traveling on company business. This per diem is paid for each full day of travel and is pro-rated at 50 percent for departure and return days. Receipts are not required against the per diem. Lodging is reimbursed separately at actual cost, capped at \$280 per night in tier-one cities and \$190 per night elsewhere. [S2]

model: mistral:7b retrieval: rerank
 sources:

[S1] expense_policy_2022.md § POL-EXP-006 "International Travel and Per Diem" (v1.4, archived, chars 2954–3513) dist=0.002
 [S2] expense_policy.md § POL-EXP-006 "International Travel and Per Diem" (v3.0, current, chars 3106–3810) dist=0.003
 [S3] expense_policy_2022.md § POL-EXP-005 "Meals and Entertainment" (v1.4, archived, chars 2431–2900) dist=0.491
 [S4] expense_policy.md § POL-EXP-005 "Meals and Entertainment" (v3.0, current, chars 2537–3052) dist=0.600

4. generation with DETECT_CONFLICTS on (warning, sources kept)
 Q: What is the international per diem for meals and incidentals?

The international per diem for meals and incidental expenses is 95 USD per day, as stated in the current version of POL-EXP-006 [S1]. The per diem is paid for each full day of travel and is pro-rated at 50 percent for departure and return days. Receipts are not required against the per diem. Lodging is reimbursed separately at actual cost, capped at 280 USD per night in tier-one cities and 190 USD per night elsewhere [S1]. Previously, the per diem was 75 USD per day [S2].

LINEAGE WARNING: 2 section(s) retrieved in more than one document version. Archived copies were not dropped.

POL-EXP-005 versions 1.4, 3.0
 [S3] expense_policy.md v3.0 current
 [S4] expense_policy_2022.md v1.4 archived
 POL-EXP-006 versions 1.4, 3.0

Disk grep shows 95 versus 75. One generated answer reports the current 95 USD and the previous 75 USD.

```

model: mistral:7b  retrieval: hybrid
sources:
  [S1] expense_policy.md § POL-EXP-006 "International Travel and Per Diem" (v3.0, current, chars 3106-3810) dist=0.967
  [S2] expense_policy_2022.md § POL-EXP-006 "International Travel and Per Diem" (v1.4, archived, chars 2954-3513) dist=0.968
  [S3] expense_policy.md § POL-EXP-005 "Meals and Entertainment" (v3.0, current, chars 2537-3052) dist=0.969
  [S4] expense_policy_2022.md § POL-EXP-005 "Meals and Entertainment" (v1.4, archived, chars 2431-2900) dist=0.969
  [S5] expense_policy.md § POL-EXP-001 "Purpose and Scope" (v3.0, current, chars 250-681) dist=0.970

sources identical across 3 and 4: True

=====
case: pre_approval
Q: What is the pre-approval threshold for a single expense?
=====

1. retrieval – was the gold chunk found?
  hybrid
  [S1] POL-EXP-004 v3.0 current expense_policy.md dist=0.967 <-- gold section
  [S2] POL-EXP-004 v1.4 archived expense_policy_2022.md dist=0.967 <-- gold section
  [S3] POL-EXP-001 v3.0 current expense_policy.md dist=0.969
  [S4] POL-EXP-001 v1.4 archived expense_policy_2022.md dist=0.969
  [S5] POL-EXP-003 v3.0 current expense_policy.md dist=0.969

  LINEAGE WARNING: 2 section(s) retrieved in more than one document version. Archived copies were not dropped
  .
    POL-EXP-001 versions 1.4, 3.0
    [S3] expense_policy.md v3.0 current
    [S4] expense_policy_2022.md v1.4 archived
    POL-EXP-004 versions 1.4, 3.0
    [S1] expense_policy.md v3.0 current
    [S2] expense_policy_2022.md v1.4 archived
    This is a corpus defect: the retriever returned contradictory policy. Prefer status=current, or remove the stale document from the source system.
  rerank
  [S1] POL-EXP-004 v1.4 archived expense_policy_2022.md dist=0.000 <-- gold section
  [S2] POL-EXP-004 v3.0 current expense_policy.md dist=0.000 <-- gold section

  LINEAGE WARNING: 1 section(s) retrieved in more than one document version. Archived copies were not dropped
  .
    POL-EXP-004 versions 1.4, 3.0
    [S2] expense_policy.md v3.0 current
    [S1] expense_policy_2022.md v1.4 archived
    This is a corpus defect: the retriever returned contradictory policy. Prefer status=current, or remove the stale document from the source system.

2. source data – grepped on disk, before any embedding
  expense_policy.md:43: exceeding 500 USD requires
  expense_policy_2022.md:46: exceeding 1,000 USD requires

3. generation with DETECT_CONFLICTS off (hybrid, eval path)
Q: What is the pre-approval threshold for a single expense?

The pre-approval threshold for a single expense is currently 500 USD, as stated in [S1]. Previously, the thresh

```

The same pattern for pre-approval: both versions are retrieved, and the files disagree.

old was 1,000 USD, as stated in [S2]. Recurring subscriptions of any value require pre-approval regardless of the individual charge amount, because the aggregate annual cost is the relevant figure, as stated in [S1].

model: mistral:7b retrieval: hybrid

sources:

[S1] expense_policy.md § POL-EXP-004 "Pre-Approval Thresholds" (v3.0, current, chars 1780-2493) dist=0.967
 [S2] expense_policy_2022.md § POL-EXP-004 "Pre-Approval Thresholds" (v1.4, archived, chars 1975-2387) dist=0.967
 [S3] expense_policy.md § POL-EXP-001 "Purpose and Scope" (v3.0, current, chars 250-681) dist=0.969
 [S4] expense_policy_2022.md § POL-EXP-001 "Purpose and Scope" (v1.4, archived, chars 249-623) dist=0.969
 [S5] expense_policy.md § POL-EXP-003 "Receipt Requirements" (v3.0, current, chars 1234-1736) dist=0.969

3b. generation with DETECT_CONFLICTS off (rerank, unfiltered)

Q: What is the pre-approval threshold for a single expense?

The pre-approval threshold for a single expense is either \$1,000 (as per the archived policy in [S1]) or \$500 (as per the current policy in [S2]). The current policy requires approval through the expense system and does not allow for retrospective approval except in cases of genuine emergencies affecting employee safety. Recurring subscriptions of any value require pre-approval regardless of the individual charge amount, because the aggregate annual cost is the relevant figure. Departmental budget holders may set lower thresholds for their own teams but may not set higher ones.

model: mistral:7b retrieval: rerank

sources:

[S1] expense_policy_2022.md § POL-EXP-004 "Pre-Approval Thresholds" (v1.4, archived, chars 1975-2387) dist=0.000
 [S2] expense_policy.md § POL-EXP-004 "Pre-Approval Thresholds" (v3.0, current, chars 1780-2493) dist=0.000

4. generation with DETECT_CONFLICTS on (warning, sources kept)

Q: What is the pre-approval threshold for a single expense?

The pre-approval threshold for a single expense is currently 500 USD, as stated in [S1]. Previously, the threshold was 1,000 USD, as stated in [S2]. Recurring subscriptions of any value require pre-approval regardless of the individual charge amount, because the aggregate annual cost is the relevant figure, as stated in [S1].

LINEAGE WARNING: 2 section(s) retrieved in more than one document version. Archived copies were not dropped.

POL-EXP-001 versions 1.4, 3.0

[S3] expense_policy.md v3.0 current

[S4] expense_policy_2022.md v1.4 archived

POL-EXP-004 versions 1.4, 3.0

[S1] expense_policy.md v3.0 current

[S2] expense_policy_2022.md v1.4 archived

This is a corpus defect: the retriever returned contradictory policy. Prefer status=current, or remove the stale document from the source system.

model: mistral:7b retrieval: hybrid

sources:

[S1] expense_policy.md § POL-EXP-004 "Pre-Approval Thresholds" (v3.0, current, chars 1780-2493) dist=0.967
 [S2] expense_policy_2022.md § POL-EXP-004 "Pre-Approval Thresholds" (v1.4, archived, chars 1975-2387) dist=0.967
 [S3] expense_policy.md § POL-EXP-001 "Purpose and Scope" (v3.0, current, chars 250-681) dist=0.969
 [S4] expense_policy_2022.md § POL-EXP-001 "Purpose and Scope" (v1.4, archived, chars 249-623) dist=0.969
 [S5] expense_policy.md § POL-EXP-003 "Receipt Requirements" (v3.0, current, chars 1234-1736) dist=0.969

sources identical across 3 and 4: True

The unfiltered reranked answer states either 1,000 USD or 500 USD.

SECTION 10

Source attribution

Every chunk carries the document name, version, status, section code, section title, and character offsets. The answer cites those sources with [S#] markers, and the sources block prints the same metadata.

```

● ((.venv) ) root@f830357d28a3:/workspace/rag_policy_assistant# python scripts/ask.py --mode hybrid \
  "How many unused PTO days can I carry into the next year?"
Q: How many unused PTO days can I carry into the next year?

Employees can carry a maximum of 5 unused PTO days into the following calendar year [S1]. These carried days must be used by 31 March, after which they lapse without payment [S1]. Unused days beyond the carryover limit are forfeited at year end and are not paid out, except where local law requires otherwise [S1].

model: mistral:7b   retrieval: hybrid
sources:
  [S1] pto_policy.md § POL-PTO-003 "Carryover and Payout" (v2.0, current, chars 1264-1834) dist=0.967
  [S2] pto_policy.md § POL-PTO-001 "Accrual" (v2.0, current, chars 188-664) dist=0.968
  [S3] pto_policy.md § POL-PTO-004 "Sick Leave" (v2.0, current, chars 1865-2680) dist=0.968
  [S4] pto_policy.md § POL-PTO-002 "Requesting Time Off" (v2.0, current, chars 704-1223) dist=0.969
○ ((.venv) ) root@f830357d28a3:/workspace/rag_policy_assistant# █

```

PTO carryover answer cites [S1]: pto_policy.md, section POL-PTO-003, version 2.0, current.

SECTION 11

Passing pipeline

GitLab CI pipeline 2879458267 ran on commit fe391e5 of feature/ingest-pipeline. All four jobs passed: install_dependencies, run_tests, check_formatting, and check_types.

#2879458267

✓ Passed Created 5 hours ago by **mani m**, finished 5 hours ago

For commit `fe391e5f`  Prefer current policy in answers unless the question needs the archive.

In `feature/ingest-pipeline`

latest branch 4 jobs 6.64 4 minutes 29 seconds, queued for 0 seconds

Pipeline	Jobs 4	Tests 0
Status	Job	Stage
✓ Passed ⌚ 00:02:44 📅 5 hours ago	#16714355362: check_types 🗑 feature/ingest-pipeline -🔗 fe391e5f	test
✓ Passed ⌚ 00:00:32 📅 5 hours ago	#16714355361: check_formatting 🗑 feature/ingest-pipeline -🔗 fe391e5f	test
✓ Passed ⌚ 00:01:36 📅 5 hours ago	#16714355360: run_tests 🗑 feature/ingest-pipeline -🔗 fe391e5f	test
✓ Passed ⌚ 00:01:44 📅 5 hours ago	#16714355359: install_dependencies 🗑 feature/ingest-pipeline -🔗 fe391e5f	build

Jobs tab: every required job passed.

#2879458267

✓ Passed Created 5 hours ago by **mani m**, finished 5 hours ago

For commit `fe391e5f`  Prefer current policy in answers unless the question needs the archive.

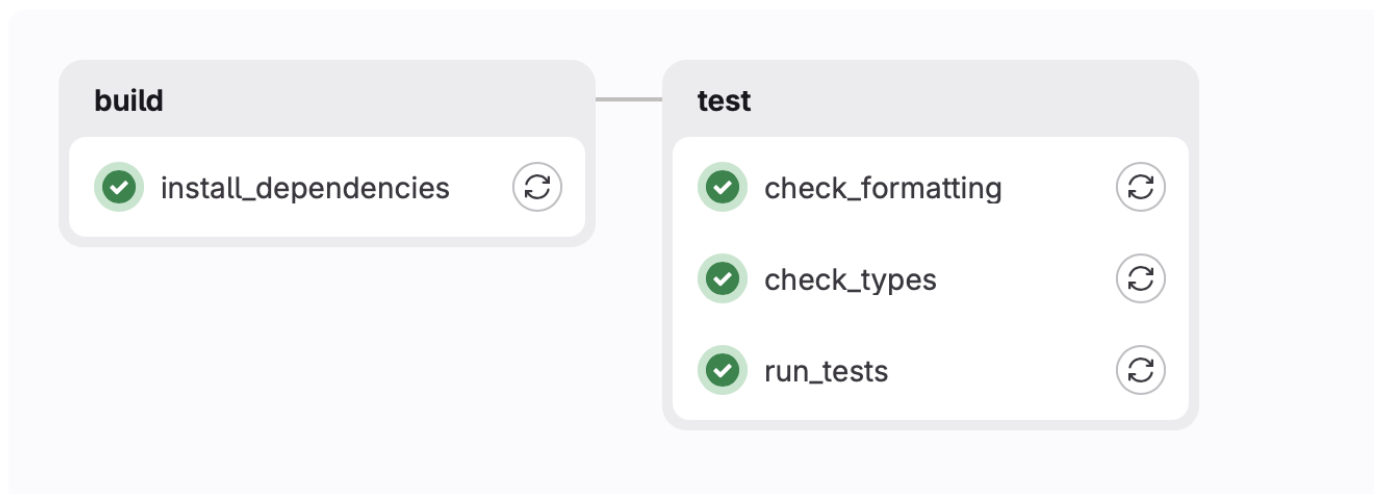
In `feature/ingest-pipeline`

latest branch 4 jobs 6.64 4 minutes 29 seconds, queued for 0 seconds

Pipeline

Jobs 4

Tests 0



Pipeline graph: build and test stages are green.